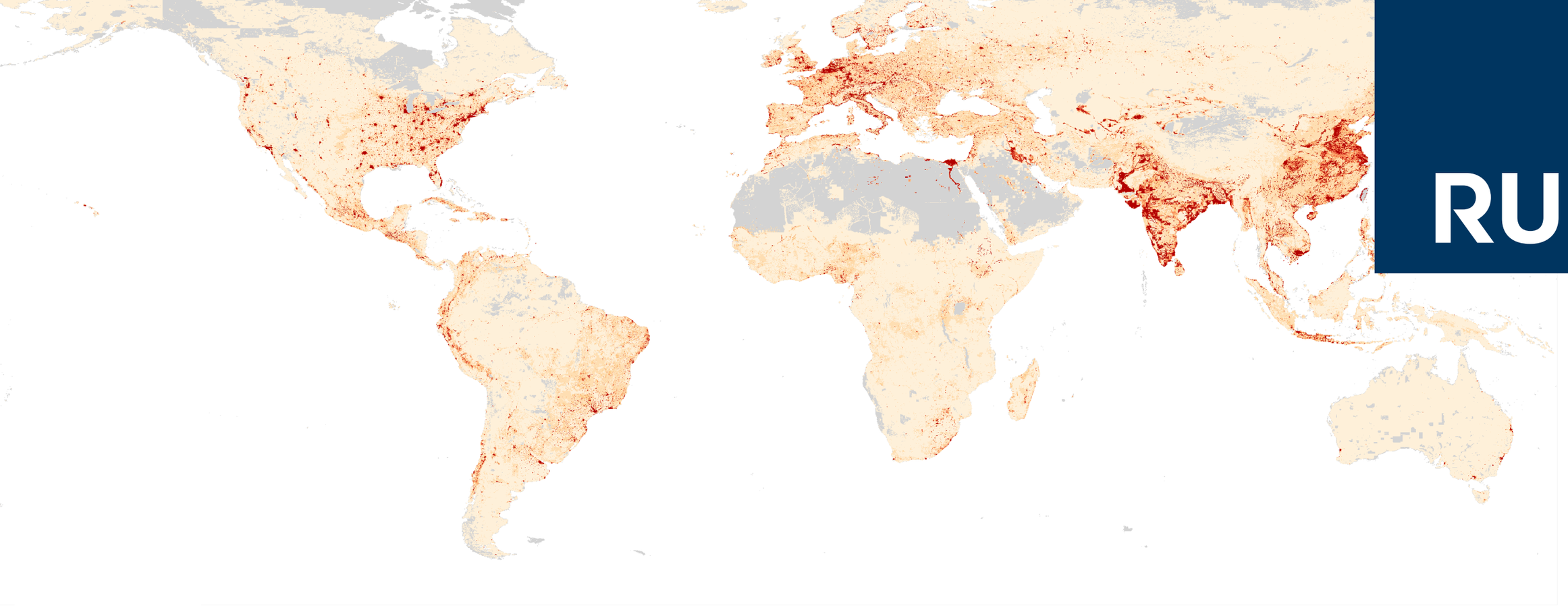


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GLOBAL CHANGE HYDROLOGY (GCH)

The WorldQual Model – A short Introduction

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Why do we need to model global water pollution?

Idea: Analysis of available data + prudent use of global modelling

- Compensate for very poor data coverage (e.g. 23 stations from GEMS for BOD in Africa) – Models provide preliminary information for data-sparse rivers
- Combine logically two kinds of information we know “fairly” well – pollution loadings & dilution capacity
- Identify current hotspot areas
- Identify priority areas for acquiring data
- Analyze effectiveness of options to reduce water pollution
- Estimate trends over next 20 years – Identify new potential hotspots; identify changing direction of water quality
- Assess impact of climate change & economic development on water quality

Global/Large-scale Water Quality Models

Table 2.2 Brief summary of the large-scale water quality models used in the current report.

Models	Simulated water quality parameters		Water body type[2]	Spatial aggregation of model outputs		Temporal aggregation of model outputs		Key references
	Parameter group	Parameters[1]		Resolution[3]	Coverage	Resolution[3]	Baseline year	
DRASTIC	Nutrients	NO_3^-	a	15 km	Africa	10-year	1990-2010	Ouedraogo *et al.* (2016)
GlobalAsGW	Geogenic contaminants	Arsenic	a	30 arcseconds	Global	NA (static)[4]	Pre-2019	Podgorski and Berg (2020)
GloWPa	Microorganisms	Cryptosporidium	b	0.5 degree	Global	Monthly	Around 2010	Vermeulen *et al.* (2019)
GREMiS	Others	Microplastics	b, d	Basin	Global	Annual	2000	van Wijnen *et al.* (2019)
IMAGE-GNM	Nutrients	TN, TP, Si	a, b, c	0.5 degree	Global and (sub-)national	Annual	1970-2015	Beusen *et al.* (2015), van Puijenbroek *et al.* (2019)
Insecticide model	Pesticides	Insecticides[5]	b	5 arcminutes	Global	NA (static)[4]	2000-2010	Ippolito *et al.* (2015)
MARINA-Global (multi-pollutant)	Nutrients	DIN, DON, DIP, DOP	b, d	Sub-basin	Global	Annual	2010	Strokal *et al.* (n.d., 2016, 2019), van Wijnen *et al.* (2017)
	Microorganisms	Cryptosporidium						
	Others	Microplastics, Triclosan						
MARINA (version 2.0)	Nutrients	DIN, DON, DIP, DOP	b, d	Sub-basin	China	Annual	2012	Wang *et al.* (2020a)
	Others	ICEP						
QUAL	Physical	Water temperature	b, c	0.5 degree	Global	Monthly	1980-2010	van Vliet *et al.* (2020)
	Organics	BOD						
	Salinity	TDS						
WaterGAP-WorldQual	Physical	Water temperature	b, c	5 arcminutes	Global	Monthly	1971-2010	Punzet *et al.* (2012)
	Nutrients	TP			Africa, Asia, Europe and Latin America		1990-2010	Voß *et al.* (2012), Reder *et al.* (2015), Fink *et al.* (2018)
	Organics	BOD						
	Salinity	TDS						
	Microorganisms	Faecal Coliform						
WFLOW-DWAQ	Others	Contaminants[6]	b, c	1 km	Europe	Annual	2017-2018	van Gils *et al.* (2020)

1. NO_3^-: nitrate, TN: total nitrogen, TP: total phosphorus, Si: Silica, BOD: biological oxygen demand, TDS: total dissolved solids, DIN: dissolved inorganic nitrogen, DON: dissolved organic nitrogen, DIP: dissolved inorganic phosphorus, DOP: dissolved organic phosphorus, ICEP: Indicator for coastal eutrophication potential.

Additional models: DynQual, mQM, GWAVA etc.

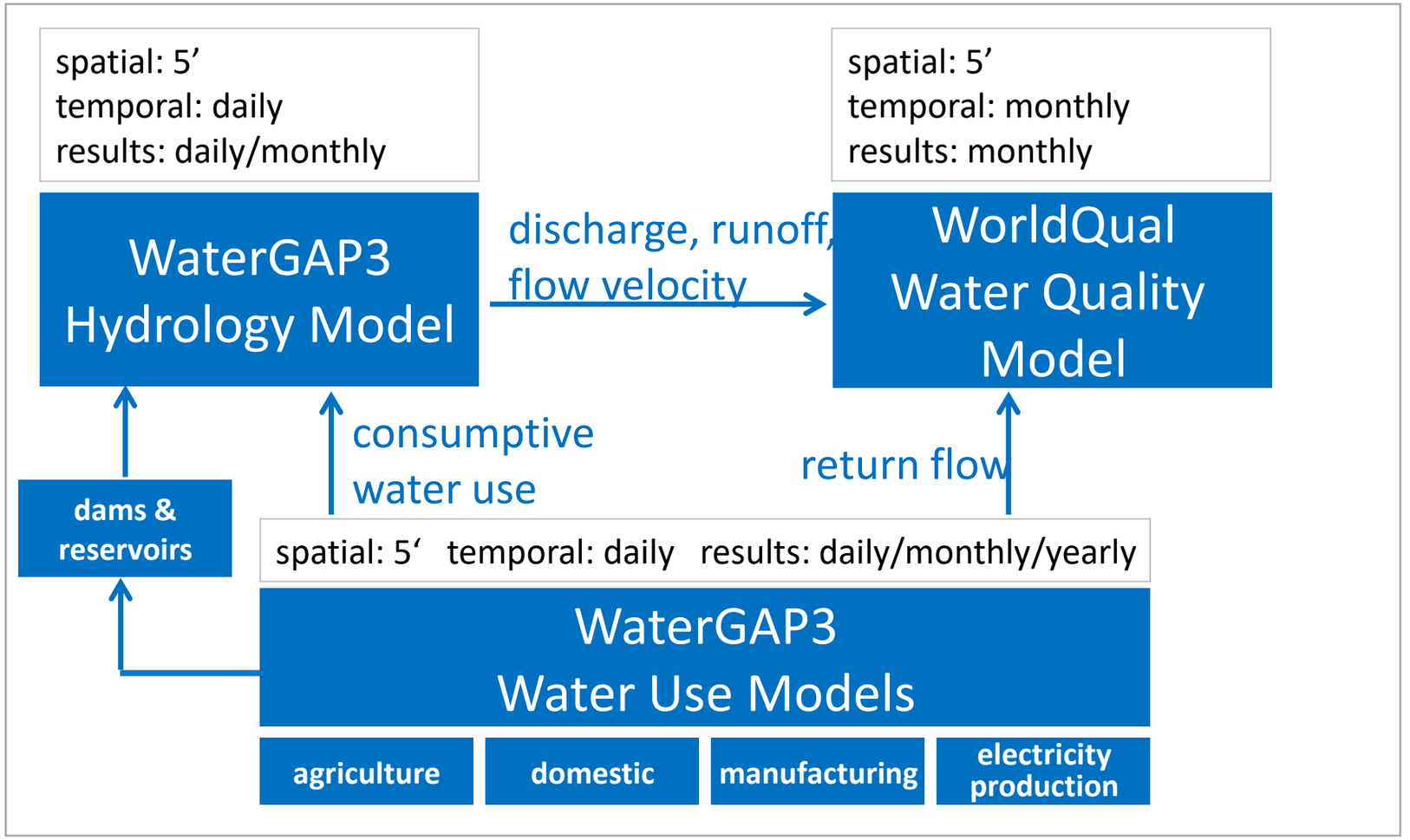
(UNEP, 2021)

L ulation resolution due to aggregation or averaging.

5. Vulnerability, hazard and risk potential for insecticide runoff, non-substance specific.
6. The model simulates the cumulative impact on ecology for 1785 chemical of emerging concern, including pharmaceuticals and pesticides.

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WaterGAP3 Modelling Framework



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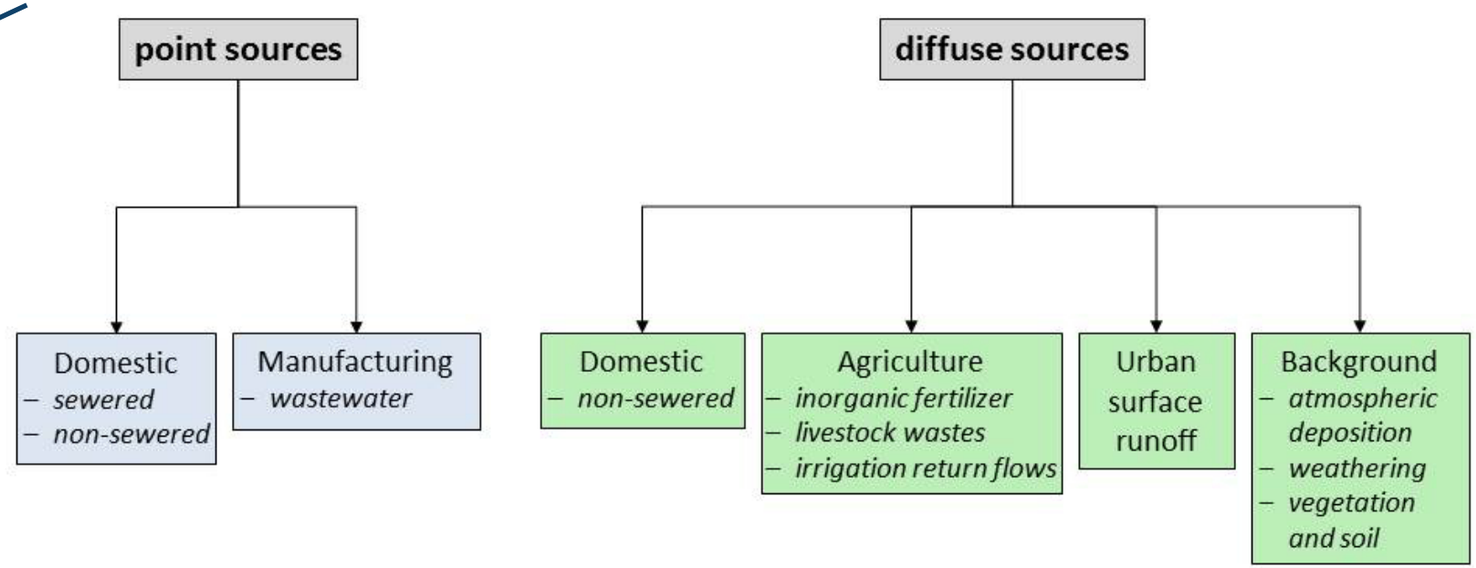
WorldQual as Part of the WaterGAP3 Modelling Framework

Parameters: FC, BOD, TDS, TP

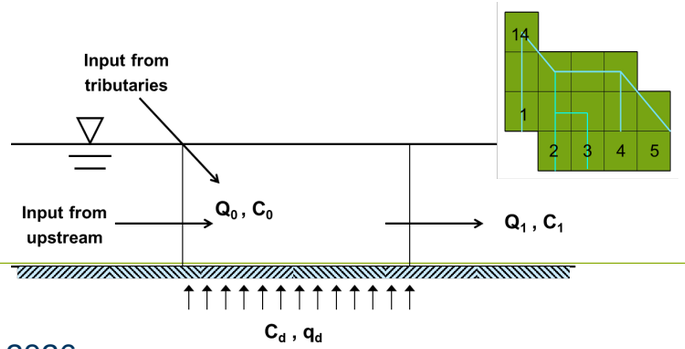
spatial: 5’
temporal: monthly
results: monthly

WorldQual Water Quality Model

Loadings calculation



Concentrations (non-)conservative substances



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About the WorldQual Model





Model calculations…

…a systems approach

Pollution loading ⇨ Dilution capacity of river + In-stream processes ⇨ Pollutant concentration in rivers

In-stream processes: decay as a function of water temperature (BOD); die-off of bacteria (FC) related to solar radiation and temperature, sedimentation of bacteria, retention (TP)

Main Substances Modelled

- **Biochemical oxygen demand (BOD5)** – [BOD directly affects the amount of dissolved oxygen in rivers; the higher the BOD, the more oxygen is depleted; less oxygen is available for aquatic organisms…fishery production]
- **Total dissolved solids (salinity)** – [High TDS results in undesirable taste; high levels could impede the functions of certain industrial applications; increased risk of salinity-associated plant toxicity]
- **Fecal coliform bacteria (FC)** – [Health risk associated with the presence of elevated levels of fecal bacteria; may indicate the presence of disease-carrying organisms]
- **Total Phosphorous (TP)** – Eutrophication is the over-fertilisation of lakes and other water bodies which leads to a disruption of their natural processes. Lake eutrophication is usually caused by anthropogenic loads of phosphorus, but loads of nitrogen can also play a role.

Sources of model uncertainty:

- Estimate of loadings: loadings per capita, treatment levels, …
- Dilution capacity: climate input, wastewater intakes, …

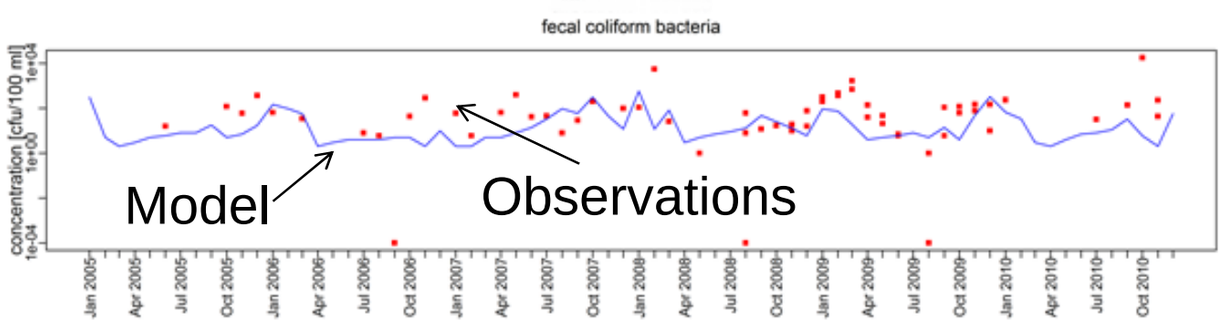
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Scientific Justification

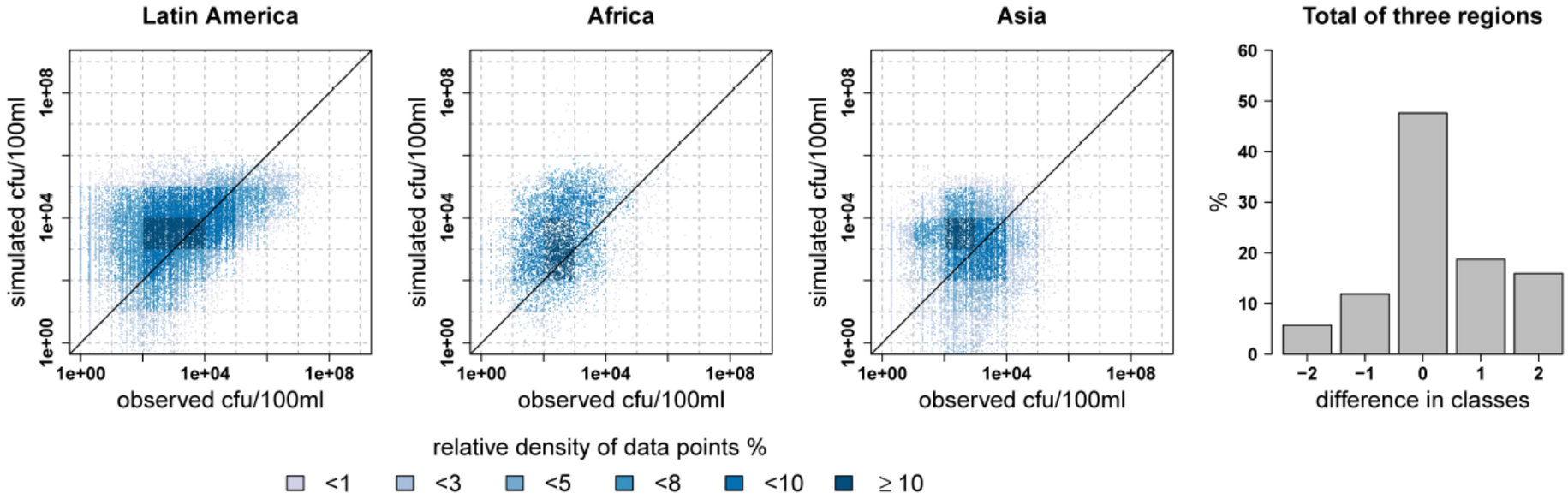
- **Based on appropriate, scientifically-accepted equations for water quality in rivers**
- **Documented in scientific literature**
- **Adapted step-by-step for developing countries:**
  - Tested and calibrated against data from rivers in Africa, Asia, Latin America
  - Pollution loadings take into account conditions in developing countries
    - Pollution caused by 8 kinds of sanitation practice
    - Loading levels per person
    - Climate and river flow conditions in developing countries
- **Used and further improved in model intercomparison studies (ISIMIP)**

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Testing the FC-Model: Model Results vs. Measurements



Station:
Vaal River near Joburg
2005-2010



Model capability to calculate correct water quality class

Source: Reder (2017)

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Limitations of Global Water Quality Modelling

- **Model only used for “suitable” questions!**
  - Where is water pollution getting better or worse, e.g. according to changes in classes?
  - What is extent and location of hotspots?
  - What are major sources of pollution leading to hotspots?
- **Is the model accurate enough to answer these suitable questions?**
- **Model part of ongoing assessment process – more data, improved model, more useful for use by stakeholders**
- **Results expressed with uncertainty range or sensitivity analysis**